

Qualidade de Software

Mestrado Integrado em Eng^a Informática – academic year 2023/2024, 1st semester

Specification for 1st report for QS2324 – version 1.0. Submission deadline: October 20th (Friday)

This is the specification for the 1st report (of two) for QS2324. Two systems of non-trivial dimensions are to be analyzed: (1) a system specific of each group selected by the students and registered at the course's wiki and (2) a system common to all groups (*wasp*) which will serve as a benchmark.

You need to carry out successful builds – for the purpose of working with the tools. If errors occur that do not prevent the tools to work, you can ignore them. You do not need to run the systems.

Format for the two reports is ACM. A doc template is available at *Moodle* (LaTeX templates can be found at the *internet*). **The report should never exceed 7 pages.** Write in either Portuguese or English.

The 1st report is intended to build an initial diagnosis of the subject systems – a "view of the forest" in preparation for the 2nd report, which will zoom in to whatever "trees" prove most problematic or worth investigating. The tools to be used for this report 1 are: (1) *SourceMiner*, (2) *inFusion* and (3) *Metrics* plugin for *Eclipse*. The concepts most centrally covered by the analyses are the ones from the book by Lanza et al up to and including the *Overview Pyramid* and the *Identity* and *Collaboration* disharmonies except *Significant Duplication*. *Classification* disharmonies may be left for the 2nd report.

The report should provide a short presentation of the specific subject system, e.g., version numbers, application domain, whether is a full-fledged application or a framework, mode of expected use (e.g., web, standalone), build tool used if any.

There is no single path to follow in the analyses. Students have much latitude in deciding how to conduct them. Nevertheless a few guidelines follow. Ideally, the first impressions and initial exploration should be carried out either using either the *Overview Pyramid* or the views of *SourceMiner*. The *Metrics* plugin is useful as a complementary source of information.

Some 80% of grading will be about the analyses and their reporting: questions asked, suitability of highlighted elements and figures, relating multiple metrics. A very important factor in the analyses and conclusions is their justification or support based on the explicitly mentioned metrics and indicators.

The analyses are much about asking interesting questions and using the tools to answer them. Some questions may arise as a reaction to something you notice and think it warrants further investigation. For the 1st report, students are not expected to take a long time looking at actual code. Nevertheless a quick look at code may be useful in some cases (i.e., something that stands out as curious or unexpected). While working on the 1st report, seize the opportunity to take notes and gather useful pointers for the subsequent analyses (2nd report, mostly focused on code smells and clones).

In case you have something useful to say about, e.g., how the project was mounted on *Eclipse*, technical obstacles overcame, or any prior "curating" that was required, describe it in a way that e.g., can be easily and quickly replicated on another system. Use a specific section for that purpose.

At most 20% of grading will be about good technical writing, report organization, a proper abstract, suitable layout of selected figures – all in view of easing report reading. Language style should be concise, precise and to the point. Do not restate theory from the lectures; just cite the (citable) sources. Report organization should be by system and not by tool, with separate sections for different topics, presented in an intuitive order. Captions for figures relating to a system need not be long but should provide the relevant information. In citations, include the standard items: author's first name (at least), title, publisher (university in case of theses and reports), year of publication. For web entries, a title, URL and date of latest access.

A summary of conclusions regarding both systems (relative strengths and weaknesses) is expected near the end.